

JavaScript Data Types

What is a Data Type?

A **Data Type** tells JavaScript what kind of value a variable stores.

```
let name = "Rokon"; // String
let age = 23; // Number
```

Types of Data

JavaScript has **2 main types** of data:

1. Primitive Data Types
2. Non-Primitive (Reference) Data Types

1. Primitive Data Types

Primitive data types store a **single value** and are **immutable**.

There are **7 primitive data types**:

Data Type	Description	Example
String	Text	"Hello"
Number	Integer or Decimal	10, 3.14
Boolean	true or false	true
Undefined	Variable declared but not assigned	undefined
Null	Intentionally empty value	null
BigInt	Very large integer	123456789n
Symbol	Unique identifier	Symbol("id")

Examples

```
let name = "Rokon";
let age = 23;
let isStudent = true;
let city;
let car = null;
let bigNumber = 12345678901234567890n;
let id = Symbol("userId");
```

2. Non-Primitive (Reference) Data Types

Non-Primitive data types can store **multiple values** and are stored **by reference**.

| Object

```
const student = {  
  name: "Rokon",  
  age: 23,  
};  
  
console.log(student.name);
```

| Array

```
const fruits = ["Apple", "Mango", "Orange"];  
  
console.log(fruits[0]);
```

| Function

```
function greet() {  
  console.log("Hello!");  
}  
  
greet();
```

| Primitive vs Non-Primitive

Feature	Primitive	Non-Primitive
Stores	Single Value	Multiple Values
Mutable	No	Yes
Copied By	Value	Reference
Examples	String, Number, Boolean	Object, Array, Function

| Primitive Example

```
let a = 10;
let b = a;

b = 20;

console.log(a); // 10
console.log(b); // 20
```

Explanation: `b` gets a copy of `a`, so changing `b` does not affect `a`.

| Non-Primitive Example

```
let person1 = {
  name: "Rokon",
};

let person2 = person1;

person2.name = "Rahim";

console.log(person1.name); // Rahim
console.log(person2.name); // Rahim
```

Explanation: Both variables point to the same object, so changing one also changes the other.

| Easy Way to Remember

- **Primitive** = Each variable has its **own copy**.
- **Non-Primitive** = Multiple variables can point to the **same object**.

| Interview Question

Q: What is the difference between Primitive and Non-Primitive Data Types?

Answer:

Primitive data types store a single value and are copied **by value**, while Non-Primitive data types store multiple values and are copied **by reference**.

JavaScript Operators

| What is an Operator?

An **Operator** is a symbol that performs an operation on one or more values (operands).

```
let sum = 10 + 5;

console.log(sum); // 15
```

| Types of Operators

1. Arithmetic Operators
2. Assignment Operators
3. Comparison Operators
4. Logical Operators
5. Increment & Decrement Operators
6. String Operator
7. Ternary Operator
8. Type Operator

| 1. Arithmetic Operators

Used for mathematical calculations.

Operator	Meaning	Example
+	Addition	10 + 5
-	Subtraction	10 - 5
*	Multiplication	10 * 5
/	Division	10 / 5
%	Modulus	10 % 3
**	Exponent	2 ** 3

```
let a = 10;
let b = 3;

console.log(a + b);
console.log(a - b);
console.log(a * b);
console.log(a / b);
console.log(a % b);
console.log(a ** b);
```

| 2. Assignment Operators

Used to assign or update values.

Operator	Example	Same As
=	x = 5	Assign value
+=	x += 2	x = x + 2
-=	x -= 2	x = x - 2
*=	x *= 2	x = x * 2
/=	x /= 2	x = x / 2
%=	x %= 2	x = x % 2

```
let x = 10;

x += 5;
console.log(x); // 15

x -= 3;
console.log(x); // 12
```

3. Comparison Operators

Used to compare two values.

Operator	Meaning	Example
==	Equal (value only)	5 == "5"
===	Strict Equal	5 === "5"
!=	Not Equal	5 != 4
!==	Strict Not Equal	5 !== "5"
>	Greater Than	10 > 5
<	Less Than	5 < 10
>=	Greater Than or Equal	5 >= 5
<=	Less Than or Equal	5 <= 10

```
console.log(10 > 5);
console.log(10 == "10");
console.log(10 === "10");
```

4. Logical Operators

Used to combine conditions.

Operator	Meaning
&&	AND
!	NOT

```
let age = 20;
let hasID = true;

console.log(age >= 18 && hasID);
console.log(age < 18 || hasID);
console.log(!hasID);
```

5. Increment & Decrement Operators

Increase or decrease a value by 1.

Operator	Meaning
++	Increment
--	Decrement

```
let count = 5;

count++;
console.log(count); // 6

count--;
console.log(count); // 5
```

6. String Operator

The `+` operator joins strings.

```
let firstName = "MD";
let lastName = "Rokonuzzaman";

console.log(firstName + " " + lastName);
```

Output:

MD Rokonzaman

| 7. Ternary Operator

A short way to write an `if...else`.

Syntax

```
condition ? trueValue : falseValue;
```

```
let age = 20;  
  
let result = age >= 18 ? "Adult" : "Minor";  
  
console.log(result);
```

| 8. Type Operator

The `typeof` operator returns the data type of a value.

```
console.log(typeof "Hello");  
console.log(typeof 100);  
console.log(typeof true);  
console.log(typeof []);  
console.log(typeof {});
```

| Operator Precedence

Operators with higher precedence execute first.

```
let result = 10 + 5 * 2;  
  
console.log(result);
```

Output:

Because `*` has higher precedence than `+`.

Summary

Category	Purpose
Arithmetic	Mathematical calculations
Assignment	Assign or update values
Comparison	Compare values
Logical	Combine conditions
Increment/Decrement	Increase or decrease by 1
String	Join strings
Ternary	Short <code>if...else</code>
Type	Check data type

Easy Way to Remember

- `+` `-` `*` `/` `%` `**` → Arithmetic
- `=` `+=` `-=` `*=` `/=` `%=` → Assignment
- `==` `===` `!=` `>` `<` `>=` `<=` → Comparison
- `&&` `||` `!` → Logical
- `++` `--` → Increment / Decrement
- `+` → String Concatenation
- `?` `:` → Ternary
- `typeof` → Type Checking

Interview Questions

Q1. What is the difference between `==` and `===`?

- `==` compares **only values** (type conversion occurs).
- `===` compares **both value and data type**.

```
console.log(5 == "5"); // true
console.log(5 === "5"); // false
```

Q2. What is the difference between `&&` and `||`?

- `&&` (AND): **All conditions must be true.**
- `||` (OR): **At least one condition must be true.**

```
console.log(true && false); // false  
console.log(true || false); // true
```